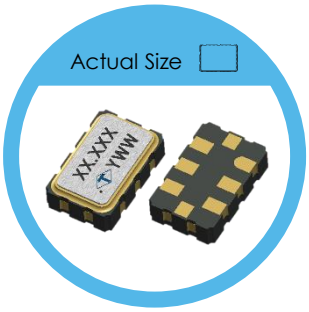
OJ-M Type High Frequency & Ultra Low Noise
5.0 x 3.2 mm Differential Output Crystal Oscillator



RoHS Compliant

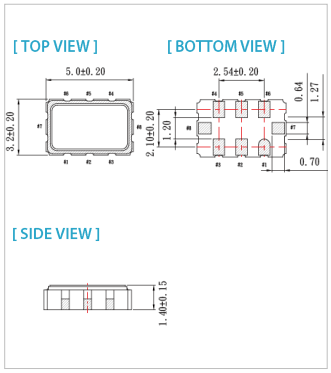
FEATURES

- Pb-free/RoHS Compliant
- Tri-state Enable / Disable Mode
- Temperature Range: -40 to 85 ˚C
- Clock Output: LVPECL, LVDS, CML and HCSL
- Output Frequency Support from 15MHz to 2.1GHz
- Low Power Supply Voltage: 3.3, 2.5, and 1.8V Supply Options
- Ultra Low Phase, RMS Jitter <300fs (Typical: 150fs at 12kHz to 20MHz Frequency Offsets)

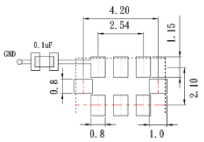
TYPICAL APPLICATION

- Set-Top Box, HDTV
- xDSL/VoIP, Cable Modem
- Jitter Attenuator, ADC

DIMENSION (mm)

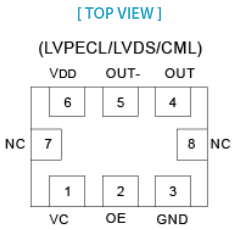


SOLDER PAD LAYOUT (mm)



Please ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads

PIN ASSIGNMENTS



PIN FUNCTION (mm)

PIN#	FUNCTION
1	NC/OE
2	OE/NC
3	GND
4	Output
5	Comp. Output
6	V_{DD}
7	NC
8	NC

ELECTRICAL SPECIFICATION

Parameter		LVPECL 3.3V Min.	LVPECL 3.3V Max.	LVPECL 2.5V Min.	LVPECL 2.5V Max.	Unit	Test Condition
Supply Voltage Variation (V_{DD}) ±10%		2.97	3.63	2.25	2.75	V	
Frequency Range		15	2100	15	2100	MHz	
Standard Frequency		100, 106.25, 125, 156.25, 187.5, 200, 212.5, 266, 300, 312.5, 400, 491.52, 622.08, 644.531250				MHz	
Supply Current			110		95	mA	
Output Level	Output High	V_{DD} – 1.165	V_{DD} – 0.8	V_{DD} – 1.165	V_{DD} – 0.8	V	
	Output Low	V_{DD} – 2.0	V_{DD} – 1.55	V_{DD} – 2.0	V_{DD} – 1.55	V	
Transition Time (20% – 80%)	Rise Time		0.35		0.35	nSec	
	Fall Time		0.35		0.35	nSec	
Duty Cycle		45	55	45	55	%	
Startup Time			8		8	mSec	
Tri-State Mode (Input to Pin 2)	Enable	0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable		0.3 x V_{DD}		0.3 x V_{DD}	V	
Stand by Current			110		95	mA	
Phase Noise		Typ.	Max.	Typ.	Max.		
At V_{DD}=3.3V, fout=873.515MHz	1kHz offset	-106		-106		dBc/Hz	
	10kHz offset	-115		-115		dBc/Hz	
	100kHz offset	-123		-123		dBc/Hz	
	1MHz offset	-133		-133		dBc/Hz	
	20MHz offset	-150		-150		dBc/Hz	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)		150	300	150	300	fs	
Period Jitter			50		50	ps	

Note: not all combination of options are available. Other specifications may be available upon request.

Specifications subject to change without notice.

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Rev(7) 04/2024

Parameter		LVDS						Unit	Test Condition
		3.3V		2.5V		1.8V			
		Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})		2.97	3.63	2.25	2.75	1.71	1.89	V	
Frequency Range		15	2100	15	2100	15	2100	MHz	
Standard Frequency		100, 106.25, 125, 156.25, 187.5, 200, 212.5, 266, 300, 312.5, 400, 491.52, 622.08, 644.531250						MHz	
Supply Current			90		80		70	mA	
Output Level	Output High		1.6		1.6		1.6	V	
	Output Low	0.9		0.9		0.9		V	
Transition Time (20% – 80%)	Rise Time		0.35		0.35		0.35	nSec	
	Fall Time		0.35		0.35		0.35	nSec	
Duty Cycle		45	55	45	55	45	55	%	
Startup Time			8		8		8	mSec	
Tri-State Mode (Input to Pin 2)	Enable	0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
Stand by Current			90		80		70	mA	
Phase Noise		Typ.	Max.	Typ.	Max.	Typ.	Max.		
At V_{DD}=3.3V, fout=873.515MHz	1kHz offset	-106		-106		-106		dBc/Hz	
	10kHz offset	-115		-115		-115		dBc/Hz	
	100kHz offset	-123		-123		-123		dBc/Hz	
	1MHz offset	-133		-133		-133		dBc/Hz	
	10MHz offset	-150		-150		-150		dBc/Hz	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)		150	300	150	300	150	300	fs	
Period Jitter			50		50		50	ps	

Parameter		HCSL						Unit	Test Condition
		3.3V		2.5V		1.8V			
		Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD}) ±10%		2.97	3.63	2.25	2.75	1.71	1.89	V	
Frequency Range		15	700	15	700	15	700	MHz	
Supply Current			115		100		94	mA	
Output Level	Output High	0.66	1.15	0.66	1.15	0.66	1.15	V	
	Output Low	0	0.15	0	0.15	0	0.15	V	
Transition Time (20% – 80%)	Rise Time		0.4		0.4		0.4	nSec	
	Fall Time		0.4		0.4		0.4	nSec	
Duty Cycle		45	55	45	55	45	55	%	
Startup Time			8		8		8	mSec	
Tri-State Mode (Input to Pin 2)	Enable	0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
Stand by Current			115		100		94	mA	
Output Load		50Ω to GND							
Phase Noise		Typ.	Max.	Typ.	Max.	Typ.	Max.		
At V_{DD}=3.3V, fout=873.515MHz	1kHz offset	-87		-87		-87		dBc/Hz	
	10kHz offset	-110		-110		-110		dBc/Hz	
	100kHz offset	-127		-127		-127		dBc/Hz	
	1MHz offset	-138		-138		-138		dBc/Hz	
	20MHz offset	-153		-153		-153		dBc/Hz	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)		150	300	150	300	150	300	fs	
Period Jitter			50		50		50	ps	

Parameter		CML						Unit	Test Condition
		3.3V		2.5V		1.8V			
		Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})		2.97	3.63	2.25	2.75	1.71	1.89	V	
Frequency Range		15	2100	15	2100	15	2100	MHz	
Standard Frequency		100, 106.25, 125, 156.25, 187.5, 200, 212.5, 266, 300, 312.5, 400, 491.52, 622.08, 644.531250						MHz	
Supply Current			90		80		70	mA	
Output Level	Output High	V_{DD} – 0.085	V_{DD}	V_{DD} – 0.085	V_{DD}	V_{DD} – 0.085	V_{DD}	V	
	Output Low	V_{DD} – 0.6	V_{DD} – 0.32	V_{DD} – 0.6	V_{DD} – 0.32	V_{DD} – 0.6	V_{DD} – 0.32	V	
Transition Time (20% – 80%)	Rise Time		0.35		0.35		0.35	nSec	
	Fall Time		0.35		0.35		0.35	nSec	
Duty Cycle		45	55	45	55	45	55	%	
Startup Time			8		8		8	mSec	
Tri-State Mode (Input to Pin 2)	Enable	0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable		0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
Stand by Current			90		80		70	mA	
Phase Noise		Typ.	Max.	Typ.	Max.	Typ.	Max.		
At V_{DD}=3.3V, fout=805.664MHz	1kHz offset	-107		-107		-107		dBc/Hz	
	10kHz offset	-117		-117		-117		dBc/Hz	
	100kHz offset	-125		-125		-125		dBc/Hz	
	1MHz offset	-135		-135		-135		dBc/Hz	
	20MHz offset	-150		-150		-150		dBc/Hz	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)		150	300	150	300	150	300	fs	
Period Jitter			50		50		50	ps	

Note: not all combination of options are available. Other specifications may be available upon request.

Specifications subject to change without notice.

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±20	±25	±30	±50
-20 ~ +70	△	○	○	○
-40 ~ +85	×	△	○	○

○: Available △:Conditional ×: Not Available

Inclusive of calibration @ 25°C ,operating temperature rrange,input Voltage variation,load variation,aging (1st year),shock,and vibration

Note: not all combination of options are available. Other specifications may be available upon request.

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